The Rule: Merge Before Truncate

GPU 0		GPU 1	
Coefficient	Pauli String	Coefficient	Pauli String
0.7e-4	X Z Y I	0.6e-4	X Z Y I

The Same Term Is Created on Two GPUs

Cutoff = 1e-4

1 Truncate on Each GPU

0.7e-4	X Z Y I	✗	*Both Below the Cutoff*
0.6e-4	X Z Y I	✗	

Term Lost: Wrong Answer

2 Merge, Then Truncate

0.7e-4 + 0.6e-4 =

1.3e-4	X Z Y I	*Above the Cutoff*

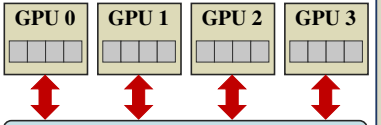
Term Kept: Correct

Measured: Kicked Ising, 22.5M Terms	No Merge		Correct Value
	4 GPUs	16 GPUs	
	0.1675	0.189	0.1531

Two Existing Ways to Merge Across GPUs

Merge Across All GPUs After Every Gate ①

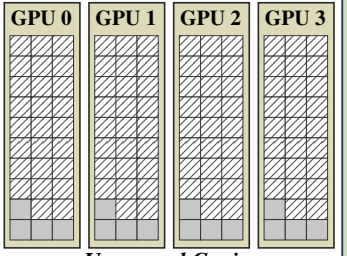
cuPauliProp Distributed Mode



Collective Calls per Gate, per GPU	22
22.5M Terms on 1 GPU	2.9 s
22.5M Terms on 16 GPUs	71 s

More GPUs, Slower

Merge Only Before Truncation ②



Unmerged Copies Stay in GPU Memory

Circuits That Run Out of Memory on 16 GPUs	2 of 7

Gives Up Communication — *One Must Be Given Up* — *Gives Up Memory*

Needed: A New Term and Its Merge Partner Always on the Same GPU

SectorQ: Merge at Every Gate, Inside a Sector

No Terms Move Between GPUs During a Window